

SEPTEMBER 10, 2019

P2.49: *Sorbitol is an ingredient in “sugar free” candy. It is sweet, but does not promote tooth decay because bacteria cannot metabolize it for food, and it is considered diet food because humans don’t metabolize it well either. Sorbitol ($C_6H_{14}O_6$) is made from glucose $C_6H_{12}O_6$ (which does cause tooth decay) and hydrogen. 100kg/day of a 30 wt% glucose solution is mixed with a stoichiometric flow rate of hydrogen and sent to a reactor; 80% of the glucose is converted to sorbitol. The hydrogen is then separated from the sugar solution as a gas stream. How much hydrogen (kg/day) is fed to the process? What is the composition (wt%) and flow rate (kg/day) of the liquid stream leaving the process?*